# Max Ellis *BDes MSc AFHEA*

## PhD Candidate in Psychoacoustics

*Psychoacoustic Assessment and Machine Learning-Based Modelling of Novel Noise Source Perception in Complex Soundscapes*

ellisacoustics.github.io · linkedin.com/in/psychoacoustic · github.com/ellisacoustics · orcid.org/0009-0007-5909-854X

## Drones, air taxis and heat pumps are about to change how our towns and cities sound forever. I work out how people will perceive them — *before* they arrive.

Noise already causes more ill health in Europe than any environmental factor except air pollution[WHO, 2011]. The next wave of technology will introduce noise that loudness alone cannot accurately judge[House of Lords, 2023]: the high-pitched whine of a rotor, the tonal hum of a compressor, rumbles of large aircraft flying low over gardens. My PhD research, sponsored by the Department for Environment, Food and Rural Affairs (Defra), seeks to accurately simulate these future soundscapes to gather response data from listening panels, for machine learning models to predict who will be annoyed, to what extent, and why. That gives regulators evidence to set standards on, and engineers a target for quieter machines, before they are everywhere.

## Research Collaborations

**Defra** — PhD Sponsor
**HEAD Acoustics GmbH** — Journal Paper
**Joby Aviation Ltd.** — Journal Paper

## Publications

2027

### Multi-Model Analysis for Prediction of Perceived Annoyance from UAVs

**M. W. Ellis**, T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, Z. Podwińska, A. J. Torija
*Journal of the Acoustical Society of America (JASA), 2027.*

Pending | Journal Paper

2027

### Psychoacoustic Assessment of UAV Noise in Varying Ambiences, Operational Conditions and Contexts

**M. W. Ellis**, Z. Podwińska, A. J. Torija
*Forum Acusticum 2027, Brussels.*

**Awarded Best Student Paper**

`Pending` `Conference Paper`

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2027

### Impacts on Soundscape Perception from eVTOL Flyovers in Existing Urban Ambiences

**M. W. Ellis**, N. Shrubsole-Green, J. Bain, Z. Podwińska, A. J. Torija
*AIAA 2027, San Diego.*

`Pending` `Conference Paper`

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2026

### Comparing a General Sound Quality Metric for Aircraft Noise Using a Pre-Trained Neural Network Model to Traditional Psychoacoustics

T. Lobato, T. Kamper-Schley, R. Sottek, M. C. Green, **M. W. Ellis**, A. J. Torija
*Quiet Drones 2026, Delft. Human Response to UAS and UAM Noise. With HEAD Acoustics GmbH.*

PDF `Conference Paper`

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2025

### Comparison of Deep Learning and Psychoacoustic Models to Predict UAV Noise Impact in Soundscapes

**M. W. Ellis**, M. C. Green, M. J. B. Lotinga, A. J. Torija Martínez
*Forum Acusticum EuroNoise 2025, 11th Convention of the European Acoustics Association, Málaga, pp. 2121–2127.*

doi:10.61782/fa.2025.0235 PDF

`Conference Paper` `Open Access`

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All publications on Google Scholar · ORCID

## Software Proficiencies

Python R MATLAB Max/MSP ArtemiS SUITE QGIS FME REAPER PyTorch TensorFlow Unity

Feature Manipulation Engine

LaTeX (Overleaf) Ableton CadnaA

## Industry Experience

Aug 2027 – Oct 2027

### Research Engineer (PhD Placement)

Dyson

- Assessing whether structural vibration can underpin sound quality metrics for product components, comparing vibration and sound quality data across anechoic, semi-reverberant and reverberant conditions.
- Applying transfer learning across sound quality metrics and jury test data from other products, to build machine learning models that predict listener response and satisfaction.

`Machine Learning` `Psychoacoustics` `Jury Testing` `DSP` `Acoustical Engineering`

Aug 2026 – Oct 2026

### Specialist Consultant (PhD Placement)

Noise Consultants Ltd. (part of Logika Group)

- Validated open-source noise measurements against modelled Defra noise-mapping data for infrastructure assessment.
- Ran a field campaign on an airport expansion project comparing physical and perceptual dominance of aircraft noise.

`Field Measurements` `Statistical Modelling` `Data Visualisation` `Acoustical Engineering`

## Academic Experience

Sep 2026 – May 2027

### Visiting Lecturer in Music Technology

University of the West of Scotland

- 1st Supervisor on the Major Project (Dissertation) module for BSc (Hons) Music Technology students specialising in Acoustics and Environmental Noise.

2026 – Present

### Associate Lecturer in Computer Science

St Mary's University, Twickenham

- Teaching, marking and moderation across the online MSc Computer Science suite.
- Module reviewer and author.

2026 – Present

### Teaching Assistant in Applied Acoustics

University of Salford

- Lab demonstrator for Environmental Noise Measurement and Modelling (BEng/MSc).
- Tutor for Immersive Sound Reproduction (MSc/PgDip).

## Education

2025 – 2028

### PhD Acoustics and Audio Engineering

University of Salford Acoustics Innovation Institute

**Year 2**

2023 – 2024

### MSc Computer Science with Artificial Intelligence

Abertay University

**Distinction**

2021 – 2023

### BDes (Hons) Sound for Moving Image

University of Glasgow

**2nd Class Honours, Upper-Division (2:1)**

2019 – 2021

### DipHE Music Technology

University of the West of Scotland

**Distinction**

**Awarded BSc Level 8 Court Medal**

2019

### BTEC Level 3 Digital Synthesis and Sampling

Fife College

**Pass**

2018 – 2019

### SCQF Level 6 Sound Production

Glasgow Clyde College Anniesland

**Pass**

## Working Groups and Committees

**NASA** Unmanned Aircraft Systems Noise Working Group (UNWG) — Member

**Noise Network Plus** A.I. and Digital Working Group — Member

**Institute of Acoustics** Scottish Branch Committee — Member; Early Careers Group — Member

## Certifications

**Advanced Prompt Engineering**

**Advanced R Programming**
Johns Hopkins University

Vanderbilt University

### AI for Noise Pollution Control
LSIB

### Applications of AI in Acoustics
European Acoustics Association Summer School

### Auralization
Autumn School Series in Acoustics (ASSA) 2026

### Digital Signal Processing
EPFL

### Ethics of Artificial Intelligence
Politecnico di Milano

### Responsible Research and Innovation
ORBITRRI

### Transformer Models and BERT Model
Google

### Trustworthy Generative AI
Vanderbilt University